

WINNER, SD, USA

WMO: 726518

Lat: 43.391N

Lon: 99.842W

Elev: 2032

StdP: 13.65

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.4	-0.3	-12.5	3.0	-4.1	-8.4	3.8	-0.1	35.6	25.3	32.9	24.2	10.6	310	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	98.6	70.8	93.8	70.4	90.5	69.6	75.5	88.9	73.9	87.3	72.2	85.8	14.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	127.6	82.9	70.0	119.0	81.5	67.9	110.5	79.8	40.5	88.5	38.8	86.9	37.2	86.1	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.3	27.1	24.6	-12.8	105.4	5.1	4.7	-16.5	108.7	-19.5	111.5	-22.3	114.1	-26.0	117.6		
			-13.3	78.2	4.9	1.4	-16.9	79.3	-19.8	80.1	-22.5	80.9	-26.1	81.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	25.2	26.4	37.7	48.1	58.5	69.3	77.3	74.3	65.9	51.2	38.8	26.6	
	DBStd	21.21	13.01	13.58	13.40	10.63	8.91	7.34	6.82	6.47	9.23	10.45	12.11	12.58	
	HDD50	3248	771	667	418	157	25	0	0	0	3	112	368	727	
	HDD65	6478	1234	1082	846	512	238	39	4	6	99	440	787	1191	
	CDD50	3278	1	5	38	101	290	580	845	755	480	150	32	1	
	CDD65	1032	0	0	1	5	38	169	384	296	126	13	0	0	
	CDH74	11560	0	1	35	119	495	1809	4439	3038	1419	188	17	0	
	CDH80	5286	0	0	6	28	160	721	2290	1415	607	57	2	0	
Wind	WSAvg	11.9	12.2	11.9	12.7	13.3	12.3	11.0	10.6	10.8	11.9	12.1	12.0	12.0	
Precipitation	PrecAvg	20.9	0.4	0.5	1.0	2.7	3.5	4.1	2.1	2.0	2.0	1.8	0.6	0.4	
	PrecMax	28.3	0.9	1.2	2.5	6.3	8.0	6.6	6.2	5.3	7.2	4.6	1.7	1.0	
	PrecMin	13.2	0.1	0.0	0.0	0.7	0.2	1.7	0.1	0.4	0.3	0.0	0.0	0.0	
	PrecStd	3.9	0.2	0.3	0.7	1.5	2.1	1.4	1.6	1.3	1.6	1.4	0.5	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	68.2	79.4	83.3	90.6	98.1	105.9	100.6	97.4	86.4	76.1	60.6	
		MCWB	44.6	49.0	54.5	57.6	66.7	72.1	71.0	70.5	67.4	61.1	53.4	45.3	
	2%	DB	52.4	58.0	71.1	77.1	84.2	91.4	99.7	96.5	91.9	79.3	67.9	52.8	
		MCWB	41.1	44.6	51.7	56.9	64.9	70.9	71.4	70.8	67.0	58.2	50.7	41.4	
	5%	DB	46.4	51.1	64.1	71.8	79.5	87.9	95.5	91.4	87.4	73.1	61.3	47.0	
		MCWB	38.1	40.5	49.4	54.2	62.4	69.2	72.2	69.8	65.8	56.1	47.0	38.5	
	10%	DB	42.8	45.3	57.3	66.1	74.7	83.7	91.2	88.1	82.1	68.1	55.4	43.0	
		MCWB	36.0	37.7	45.2	51.4	60.3	68.0	70.8	69.2	63.7	53.6	44.1	35.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	50.5	59.1	61.1	70.0	76.3	78.3	76.6	72.3	64.1	55.4	46.0
			MCDB	58.6	65.3	68.3	77.3	85.3	89.7	91.6	90.8	86.8	79.5	71.5	58.4
2%		WB	41.5	45.5	53.3	58.0	67.0	73.6	76.3	74.4	70.1	60.9	51.4	41.9	
		MCDB	51.3	56.9	67.8	74.7	80.2	86.6	89.7	87.8	85.2	74.1	65.8	51.5	
5%		WB	38.8	41.1	49.8	55.2	64.4	71.4	74.7	72.6	68.1	57.8	48.2	38.9	
		MCDB	46.2	51.1	63.4	68.9	77.1	84.4	88.2	85.8	82.2	70.5	60.9	46.8	
10%		WB	36.1	37.7	46.2	52.7	61.5	69.5	73.1	71.0	66.0	54.6	44.7	36.1	
		MCDB	42.1	44.7	57.5	64.1	72.5	81.3	86.9	84.0	79.2	66.9	55.3	42.5	
Mean Daily Temperature Range		5% DB	MDBR	18.8	20.3	21.5	22.3	21.8	22.4	24.2	23.4	24.1	22.3	21.2	17.6
			MCDBR	25.1	28.0	31.7	31.8	28.3	27.3	28.4	28.2	29.2	30.7	28.9	23.7
	MCWBR		16.5	16.7	16.9	16.3	14.2	12.1	9.8	10.6	11.3	15.3	16.0	15.5	
	5% WB	MCDBR	24.1	26.7	28.8	28.5	25.0	23.8	24.4	24.4	24.8	27.7	28.3	22.6	
		MCWBR	16.1	16.3	16.5	15.6	13.7	12.4	11.1	11.0	11.4	15.0	16.1	15.3	
Clear-Sky Solar Irradiance	taub	0.261	0.273	0.296	0.341	0.357	0.372	0.391	0.393	0.349	0.314	0.276	0.263		
	taud	2.523	2.505	2.496	2.363	2.406	2.395	2.371	2.346	2.440	2.508	2.534	2.507		
	Ebn at Noon	284	298	302	293	290	285	278	273	279	278	275	272		
	Edh at Noon	21	26	30	37	36	37	37	37	31	26	21	20		
All-Sky Solar Radiation	RadAvg	618	916	1251	1572	1868	2115	2183	1839	1479	998	678	511		
	RadStd	48	67	105	108	172	149	83	131	99	115	59	38		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page